Assignment 3:
Simulation of Steady Lid-Driven Cavity Flow Using the SIMPLE Algorithm on a 161×161 Grid

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1. Problem descriptions

Governing Equation : Navier-Stokes Equations

$$\oint_{\partial\Omega} d\vec{a} \cdot \vec{v} = 0 \qquad (1.1)$$

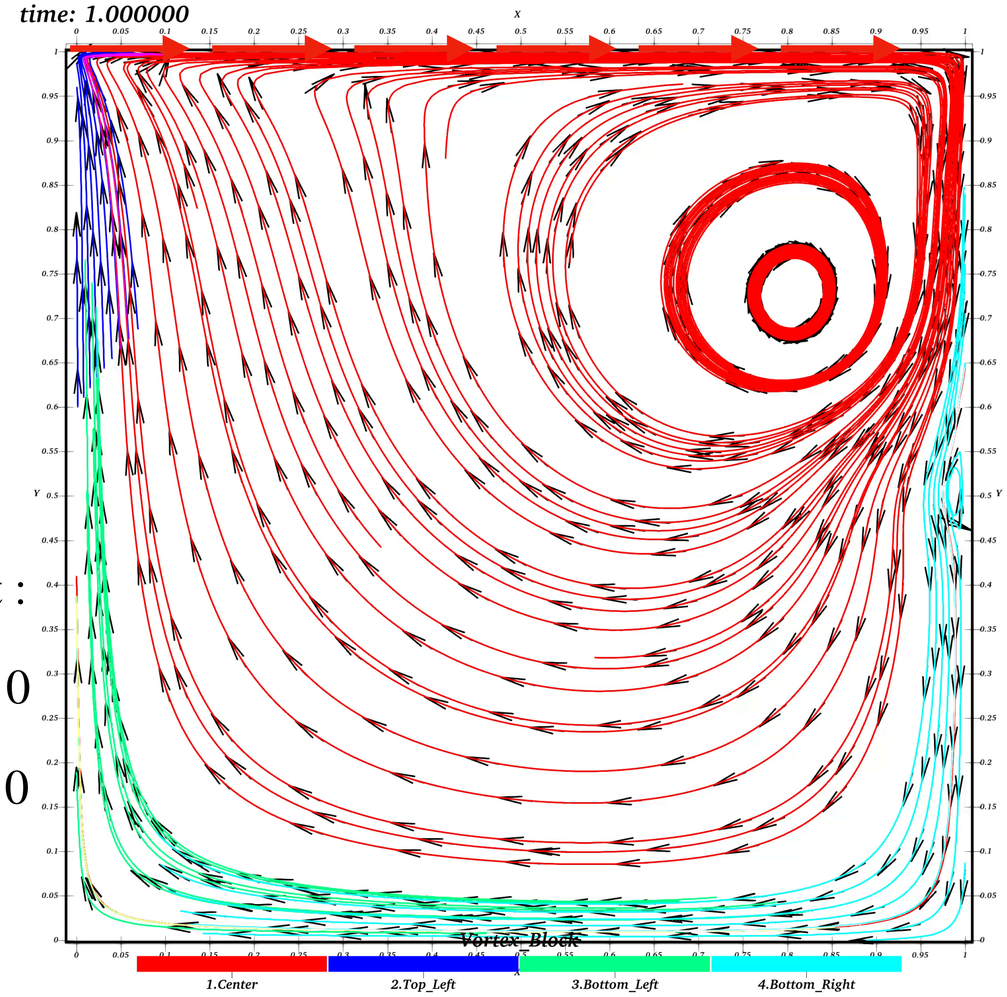
$$\oint_{\partial\Omega} d\vec{a} \cdot \left(\rho\vec{u} \otimes \vec{u}\right) = \oint_{\partial\Omega} d\vec{a} \cdot -p\overleftrightarrow{I} + \oint_{\partial\Omega} d\vec{a} \cdot \mu\overrightarrow{\nabla}\vec{u} \qquad (1.2)$$

(1) Show velocity vector field and streamlines

(2) Compare centerline velocity profiles with different Re numbers

(3) Compare centerline velocity profiles with different discrete schemes

Top : $u = 1$, $v = 0$



Left : $u = 0$, $v = 0$

Right : $u = 0$, $v = 0$

Bottom : $u = 0$, $v = 0$

▲ Fig. 1.1 Lid-Driven Cavity at Re = 5000

2. SIMPLE algorithm

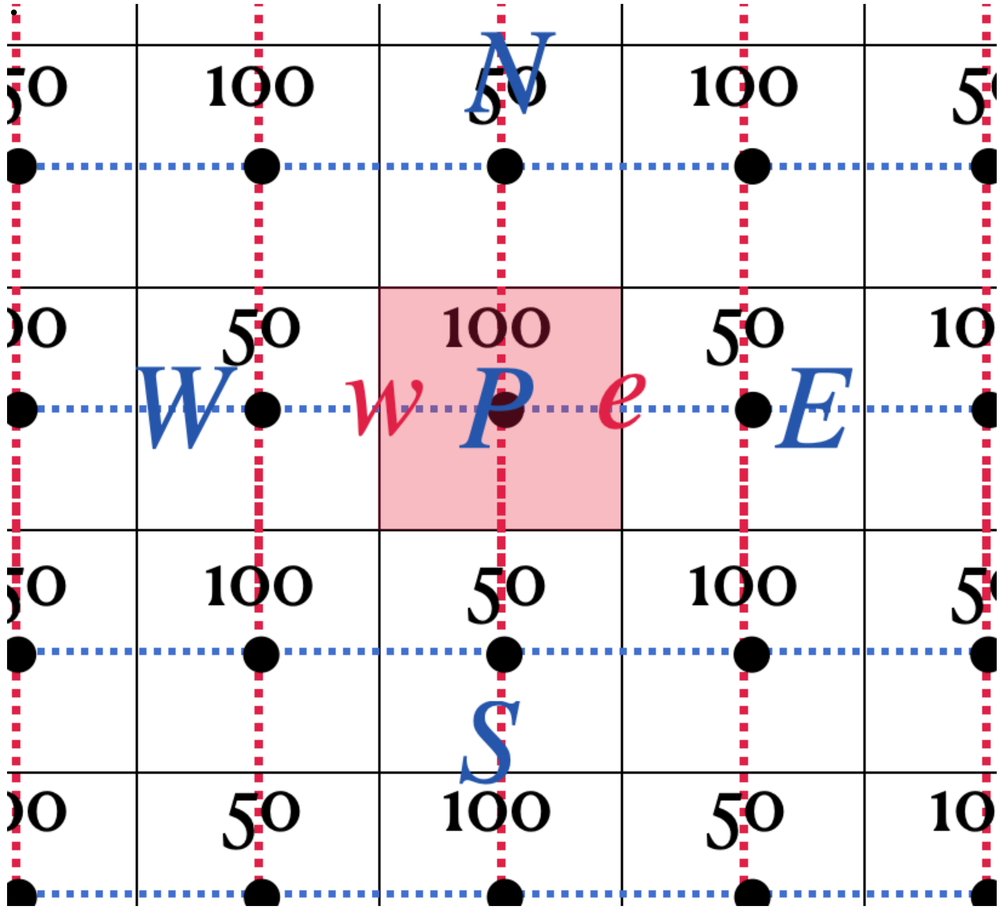
(1)staggered grid : why ? Non-physical

To understand the reason , let revisit the discretization process :

$$-\frac{1}{\rho}\oint_{\partial\Omega} dy \cdot p \approx -\frac{1}{\rho}(p_e - p_w) \approx -\frac{1}{2\rho}(p_E - p_W) \qquad (2.1)$$

Central Difference scheme

- 2-δx difference distance pressure gradient would not correct the pressure sawtooth field by momentum equation. So we use staggered grid to avoid this problem (method : 1-δx difference distance about pressure gradient), and correct checkerboard pressure field by outer iteration.



▲ Fig. 2.2 Let discuss why using the staggered grid

2. SIMPLE algorithm

(2) Problem

Discrete governing equation : How can we get the correct pressure value ?

Continuity Equation : $F_e^p - F_w^p + F_n^p - F_s^p = 0$ (2.2)

X-momentum Equation : $a_P u_P + a_{EE}u_{EE} + a_E u_E + a_W u_W + a_{WW}u_{WW} + a_{NN}u_{NN} + a_N u_N + a_S u_S + a_{SS}u_{SS}$

$$= \frac{1}{\rho}\frac{-(p_P - p_w)}{\delta x}\delta x \delta y \quad (2.3)$$

(3) Method: Predictor-Corrector Approach

Correct value = Predicted value + Correction value $p = p' + p^\star$

(i) Advantage : The pressure correction equation can be used to update the pressure correction field independently, and this equation is equivalent to the continuity equation.

(ii) Disadvantage : The number of variables increases from 3 to 9.

$$[u', v', p'] \longrightarrow [u, v, p, u', v', p', u^*, v^*, p^*]$$

2. SIMPLE algorithm

(4) Equations : Three main equations in predict-correction method

(A) Pressure Correction Equation (Pressure Poisson Equation) :

$$\left(\frac{\delta y}{\rho(a_P)^{*\,N+\frac{1}{2}}_{i+1,J}}+\frac{\delta y}{\rho(a_P)^{*\,N+\frac{1}{2}}_{i,J}}+\frac{\delta x}{\rho(b_P)^{*\,N+\frac{1}{2}}_{I,j+1}}+\frac{\delta x}{\rho(b_P)^{*\,N+\frac{1}{2}}_{I,j}}\right)p'^{\,N+1}_{I,J}$$

$$=\left(\frac{\delta x}{\rho(b_P)^{*\,N+\frac{1}{2}}_{I,j}}\right)p'^{\,N}_{I,J-1}+\left(\frac{\delta x}{\rho(b_P)^{*\,N+\frac{1}{2}}_{I,j+1}}\right)p'^{\,N}_{I,J+1}+\left(\frac{\delta y}{\rho(a_P)^{*\,N+\frac{1}{2}}_{i,J}}\right)p'^{\,N}_{I-1,J}+\left(\frac{\delta y}{\rho(a_P)^{*\,N+\frac{1}{2}}_{i+1,J}}\right)p'^{\,N}_{I+1,J}$$

$$+u^{*\,N+\frac{1}{2}}_{i,J}-u^{*\,N+\frac{1}{2}}_{i+1,J}+v^{*\,N+\frac{1}{2}}_{I,j}-v^{*\,N+\frac{1}{2}}_{I,j+1}=0 \qquad (2.4)$$

Meaning : Given the predicted velocity, obtain the predicted pressure.

$[u^*, v^*]$ → Eq. (3.3) → $[p']$

2. SIMPLE algorithm

(4) Equations : Three main equations in predict-correction method

(B) predicted momentum equation :

$$(a_P)_{i,J}^{*\,N} u_{i,J}^{*\,N+\frac{1}{2}} + (a_{WW})_{i,J}^{*\,N} u_{i-2,J}^{*\,N} + (a_W)_{i,J}^{*\,N} u_{i-1,J}^{*\,N} + (a_E)_{i+1,J}^{*\,N} u_{i+1,J}^{*\,N} + (a_{EE})_{i+2,J}^{*\,N} u_{i+2,J}^{*\,N}$$
$$+(a_{SS})_{i,J}^{*\,N} u_{i,J-2}^{*\,N} + (a_S)_{i,J}^{*\,N} u_{i,J-1}^{*\,N} + (a_N)_{i,J}^{*\,N} u_{i,J+1}^{*\,N} + (a_{NN})_{i,J}^{*\,N} u_{i,J+2}^{*\,N}$$
$$= \frac{-\left(p_{I,J}^{*\,N} - p_{I-1,J}^{*\,N}\right)}{\rho \delta x} \delta x \delta y \qquad (2.5)$$

Meaning : Given the predicted pressure , obtain the predicted velocity .

$[p^*]$ → Eq. (3.4) → $[u^*, v^*]$

Challenge : boundary treatment of the predicted form of the discrete momentum equation

2. SIMPLE algorithm

(4) Equations : Three main equations in predict-correction method

(C) Correct momentum equation :

$$(a_P)_{i,J}^{*\,N+\frac{1}{2}} u_{i,J}^{\prime\,N+1} = \frac{-\left(p_{I,J}^{\prime\,N+1} - p_{I-1,J}^{\prime\,N+1}\right)}{\rho\delta x}\delta x\delta y \tag{2.6}$$

Meaning: Given the pressure correction field, calculate the velocity correction field.

$[p']$ → Eq. (3.5) → $[u', v']$

Point about equation (3.5) : Equation (3.5) would originally take a complex form similar to Equation (3.4). However, in the SIMPLE algorithm, Patankar introduced a crucial simplification by omitting the contributions from neighboring grid points.

3. Boundary Treatment

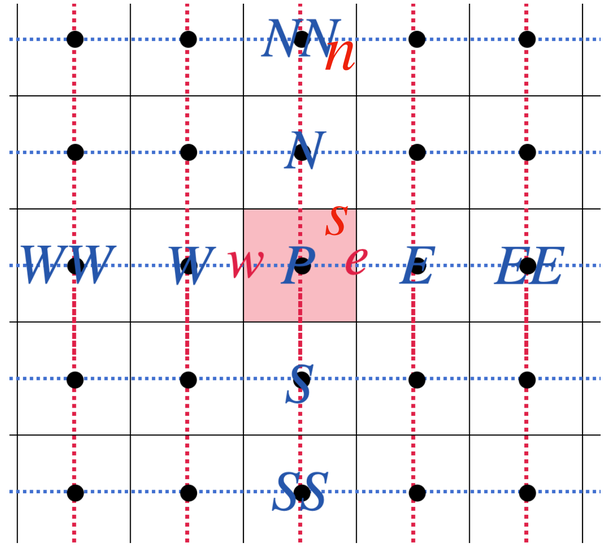
- **Governing Equation : predicted momentum equation in x- direction**

$$a_P u_P^* + a_{EE} u_{EE}^* + a_E u_E^* + a_W u_W^* + a_{WW} u_{WW}^* + a_{NN} u_{NN}^* + a_N u_N^* + a_S u_S^* + a_{SS} u_{SS}^* = \frac{1}{\rho} \frac{-\left(p_P - p_w\right)}{\delta x} \delta x \delta y \qquad (3.1)$$

$$\underbrace{F_e^u u_e^* - F_w^u u_w^* + F_n^u u_n^* - F_s^s u_s^*}_{\text{Convection term}} = \frac{1}{\rho} \frac{-\left(p_{I,J} - p_{I-1,J}\right)}{\delta x} \delta x \delta y \qquad (3.2)$$

$$\underbrace{-u_P^* \left(D_w + D_e + D_s + D_n\right) + u_W^* \left(D_w\right) + u_E^* \left(D_e\right) + u_S^* \left(D_s\right) + u_N^* \left(D_n\right)}_{\text{Diffusion term}}$$

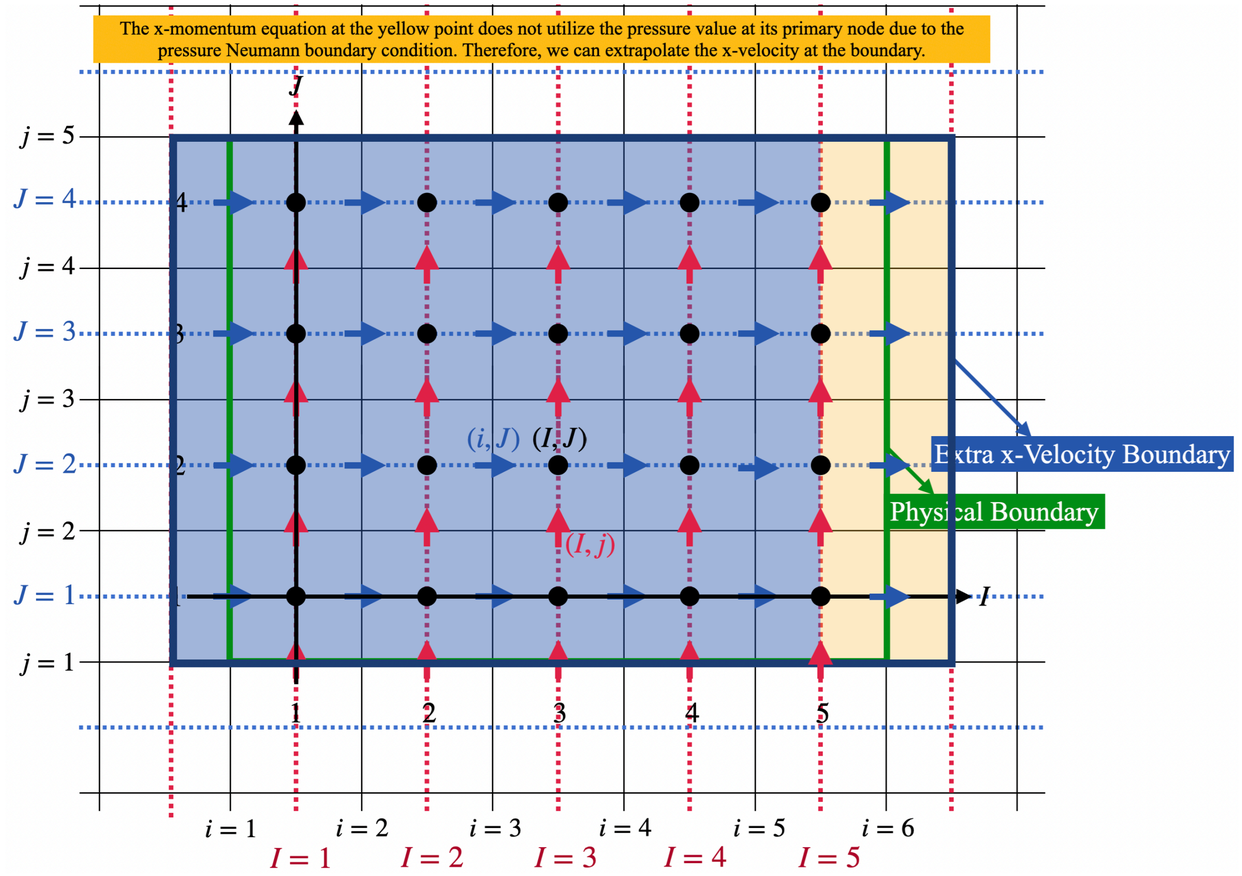
- To simplify the implementation of the four Dirichlet boundary conditions, I apply a side-by-side processing strategy instead of considering the four sides at the same time. The advantage of that is avoid complex corner treatment.



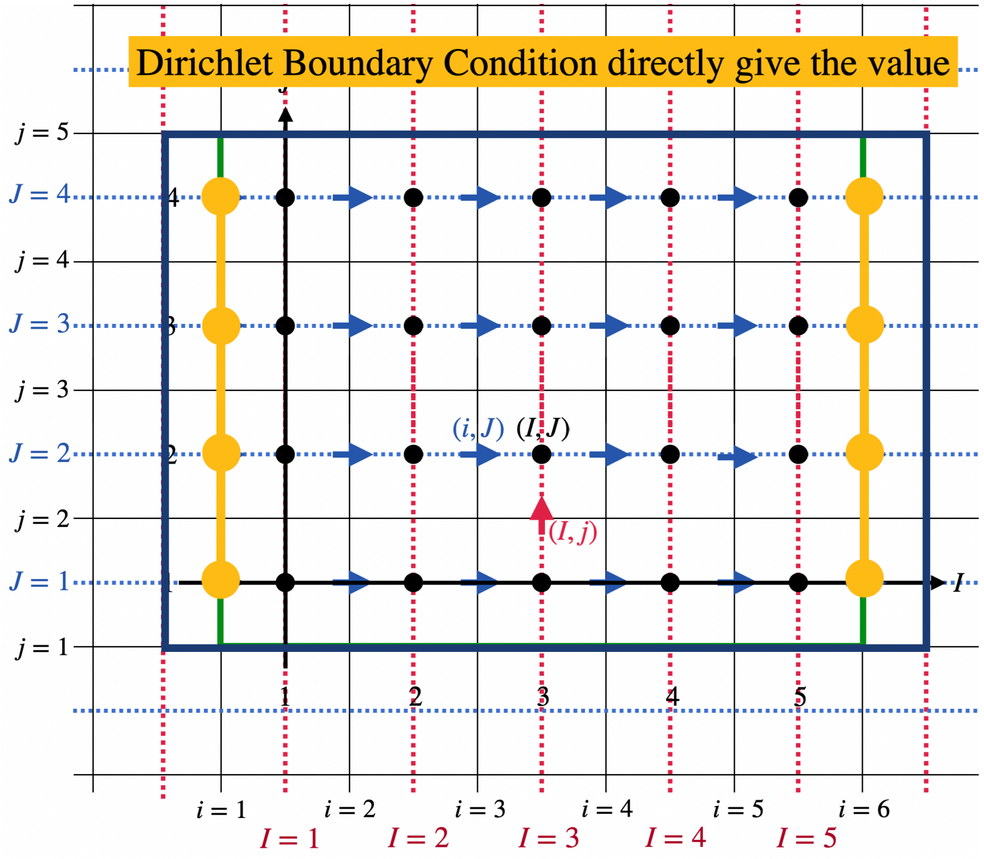
▲ Fig. 3.1 Using 8 values to decide the main point value

3. Boundary Treatment

- Part I. "The boundary conditions directly prescribe values at the primary grid points on the boundary”



▲ Fig. 3.2 u-velocity domain



▲ Fig. 3.3 For grid i = 1 and i = 6, the value of the point is directly given from boundary conditions

3. Boundary Treatment

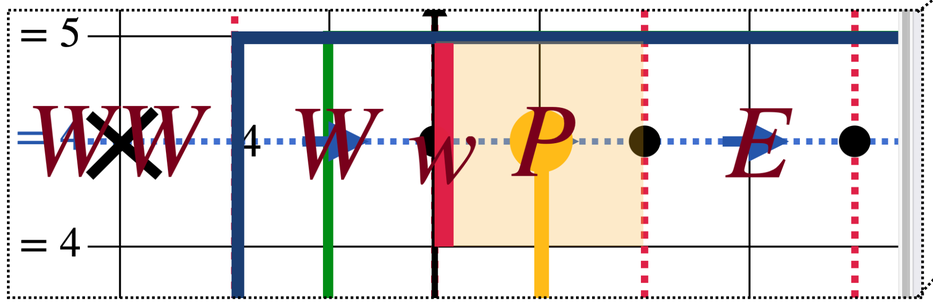
- Part II. Considering each side and applying boundary conditions : West side

The source of the boundary treatment :

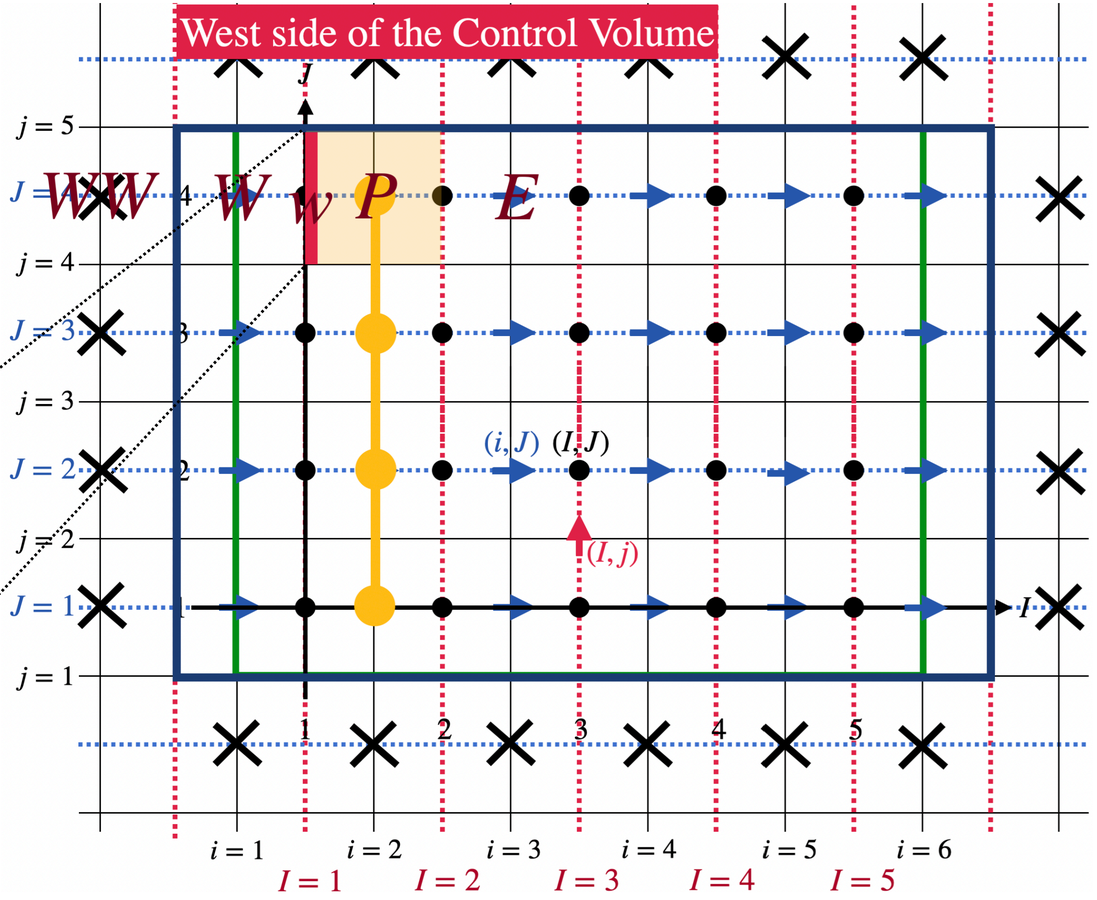
① Grid : i = 1, i.e., second left boundary points

② Convection Term : using Upwind scheme but reducing the accuracy in space.

③ Diffusion Term : maintain 2-order Central Difference



▲ Fig. 3.6 The control volume of the second left boundary points



▲ Fig. 3.5 For west side treatment, i = 2 grids require special consideration

3. Boundary Treatment

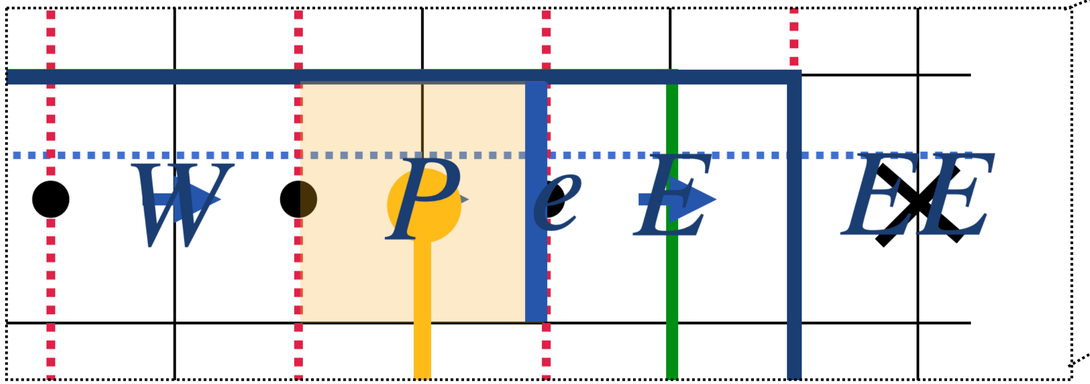
- Part II. Considering each side and applying boundary conditions : East side

The source of the boundary treatment :

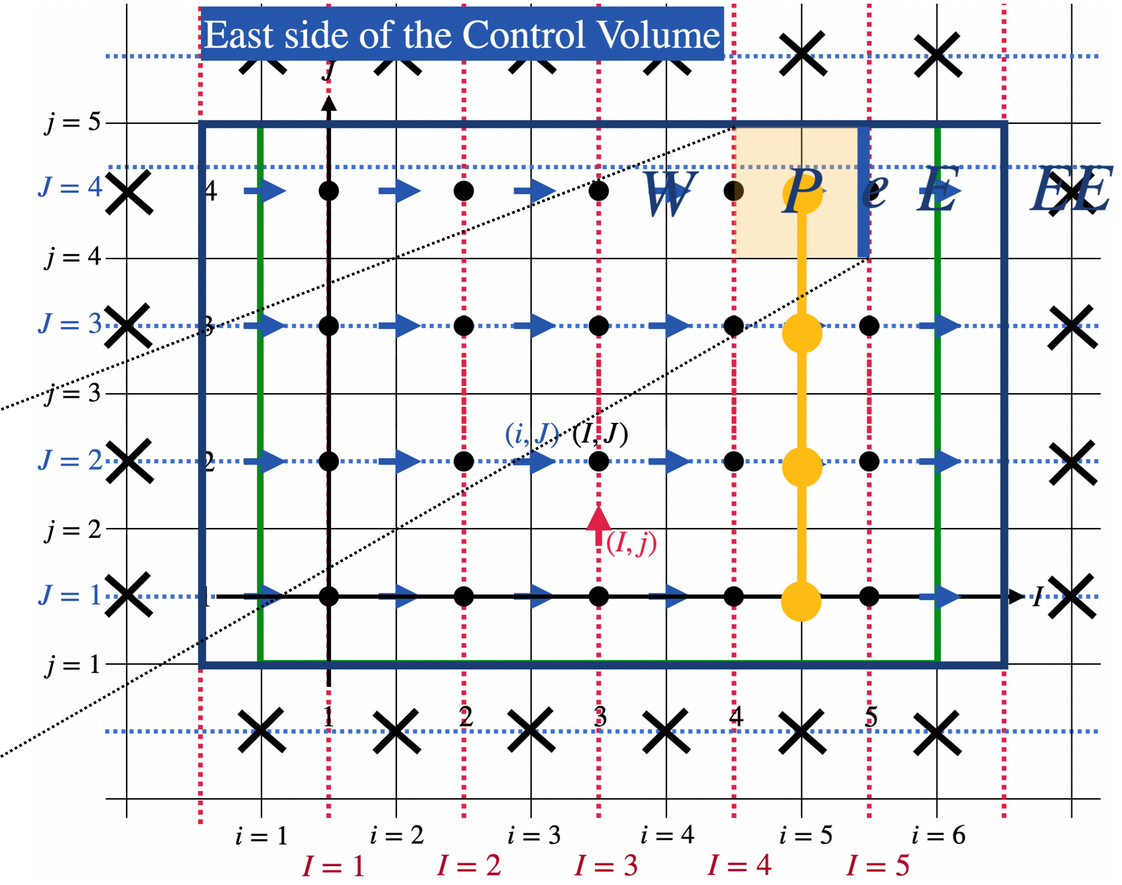
① Grid : i = Nx, i.e., second right boundary points

② Convection Term : using Upwind scheme but reducing the accuracy in space.

③ Diffusion Term : maintain 2-order Central Difference



▲ Fig. 3.8 The control volume of the second right boundary points



▲ Fig. 3.9 For east side treatment, i = Nx grids require special consideration

3. Boundary Treatment

- Part II. Considering each side and applying boundary conditions : South side

The source of the boundary treatment :

① Grid : j = 1, i.e., bottom boundary points

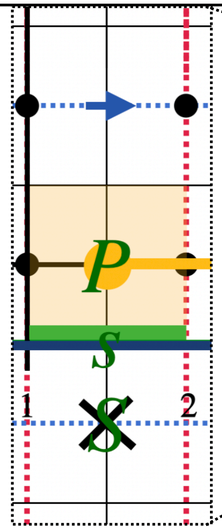
② Convection Term : directly given by Dirichlet Condition

③ Diffusion Term : maintain 2-order Central Difference but using extrapolation

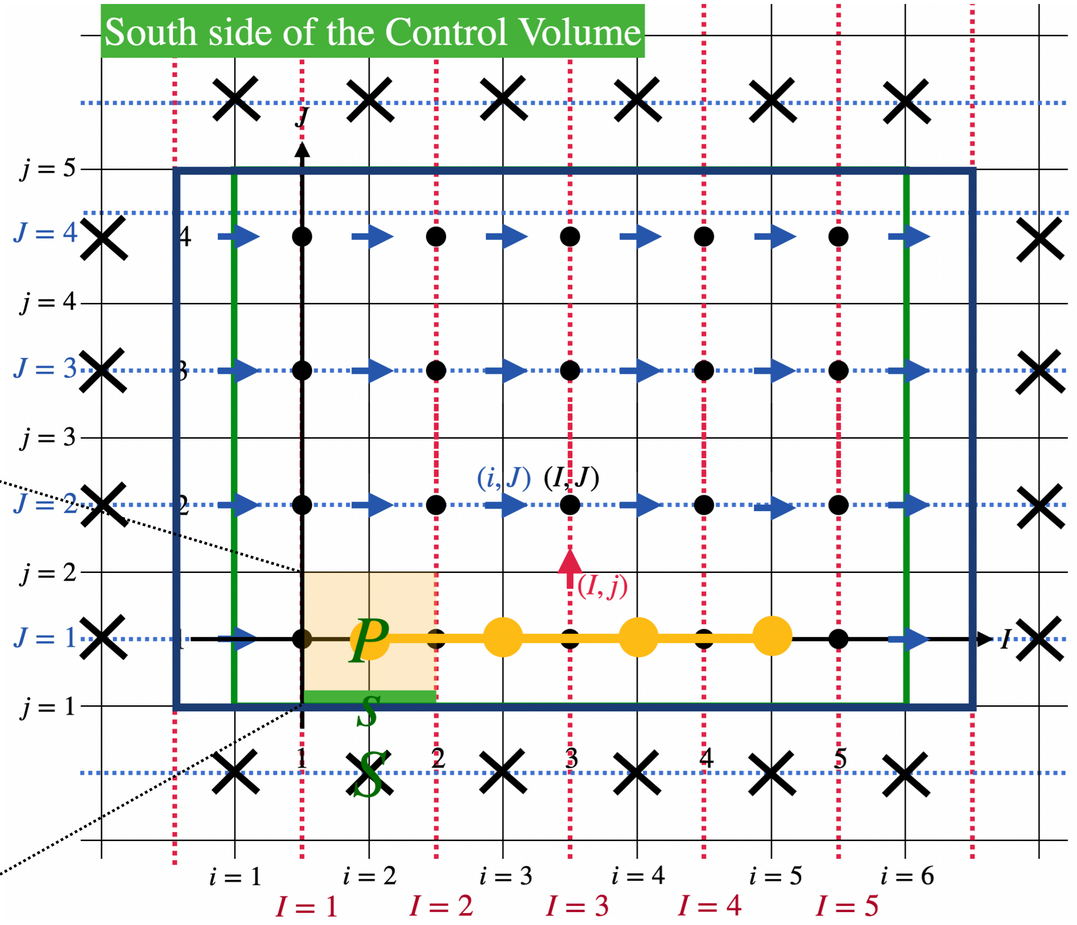
Treatment of the diffusion term :

$$-D_s\left(u_P-u_S\right)=-D_s\left(u_P-(2u_{bottom}-u_P)\right)$$

$$=-2D_s\left(u_P-u_{bottom}\right)$$



▲ Fig. 3.10 The control volume of the bottom boundary points



▲ Fig. 3.11 For south side treatment, j = 1 grids require special consideration

3. Boundary Treatment

- Part II. Considering each side and applying boundary conditions : South side

The source of the boundary treatment :

① Grid : j = 2 , i.e., second bottom boundary points

② Convection Term : Using extrapolation to obtain values at points where they are not directly available i.e., *SS* value

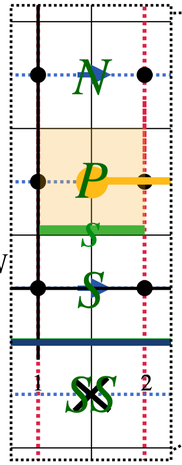
③ Diffusion Term : maintain 2-order Central Difference

Treatment of the convection term :

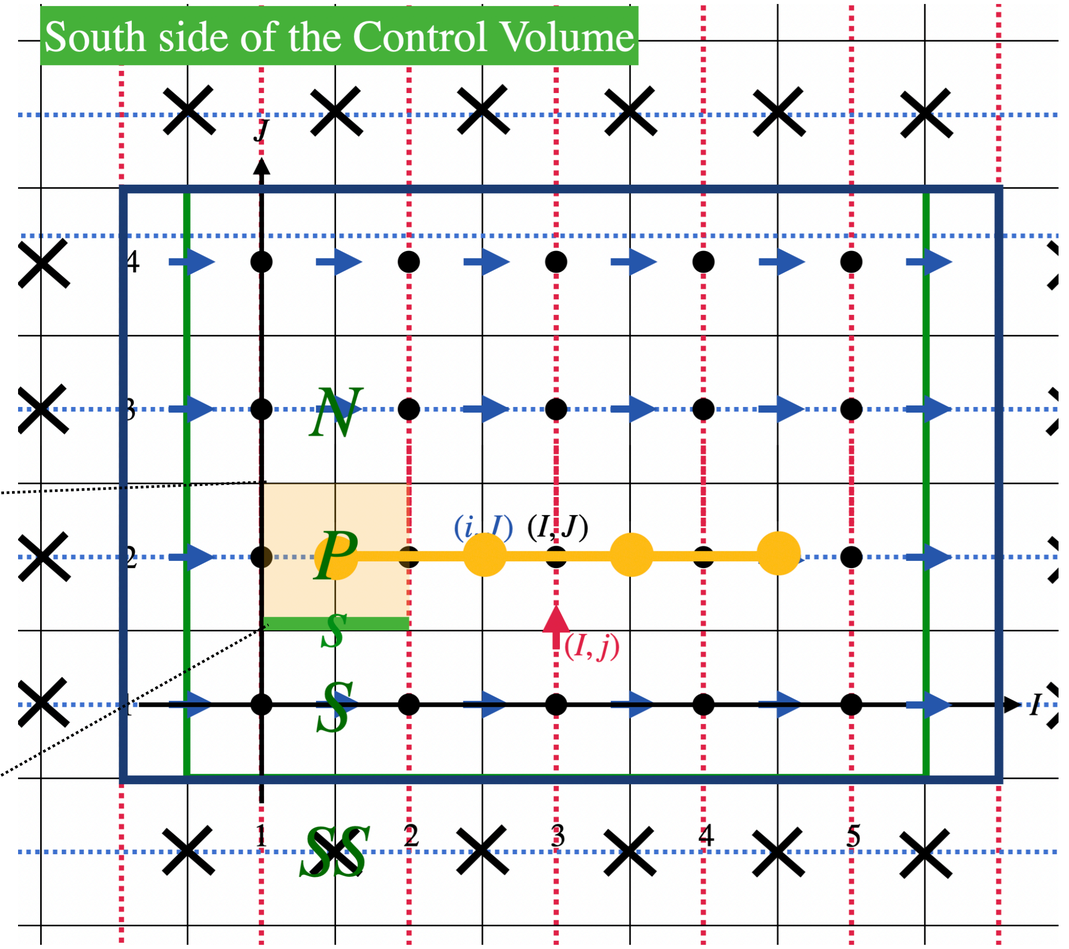
$$u_s = \left(\alpha_s UU(r_s)\right) u_{SS} + \left((1-\alpha_s)D(r_s) + \alpha_s U(r_s)\right) u_S$$

$$+\left((1-\alpha_s)U(r_s) + \alpha_s D(r_s)\right) u_P + \left((1-\alpha_s)UU(r_s)\right) u_N$$

Where, $u_{SS} = 2u_{bottom} - u_S$



▲ Fig. 3.12 The control volume of the second bottom boundary points



▲ Fig. 3.13 For south side treatment, j = 2 grids require special consideration

3. Boundary Treatment

- Part II. Considering each side and applying boundary conditions : South side

The source of the boundary treatment :

① Grid : j = Ny , i.e., top boundary points

② Convection Term : *N* point didn't exist and using extrapolation to evaluate the value.

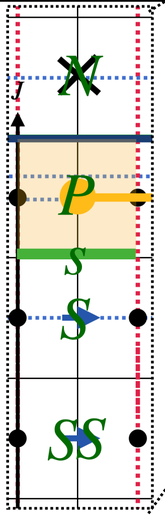
③ Diffusion Term : maintain 2-order Central Difference

Treatment of the convection term :

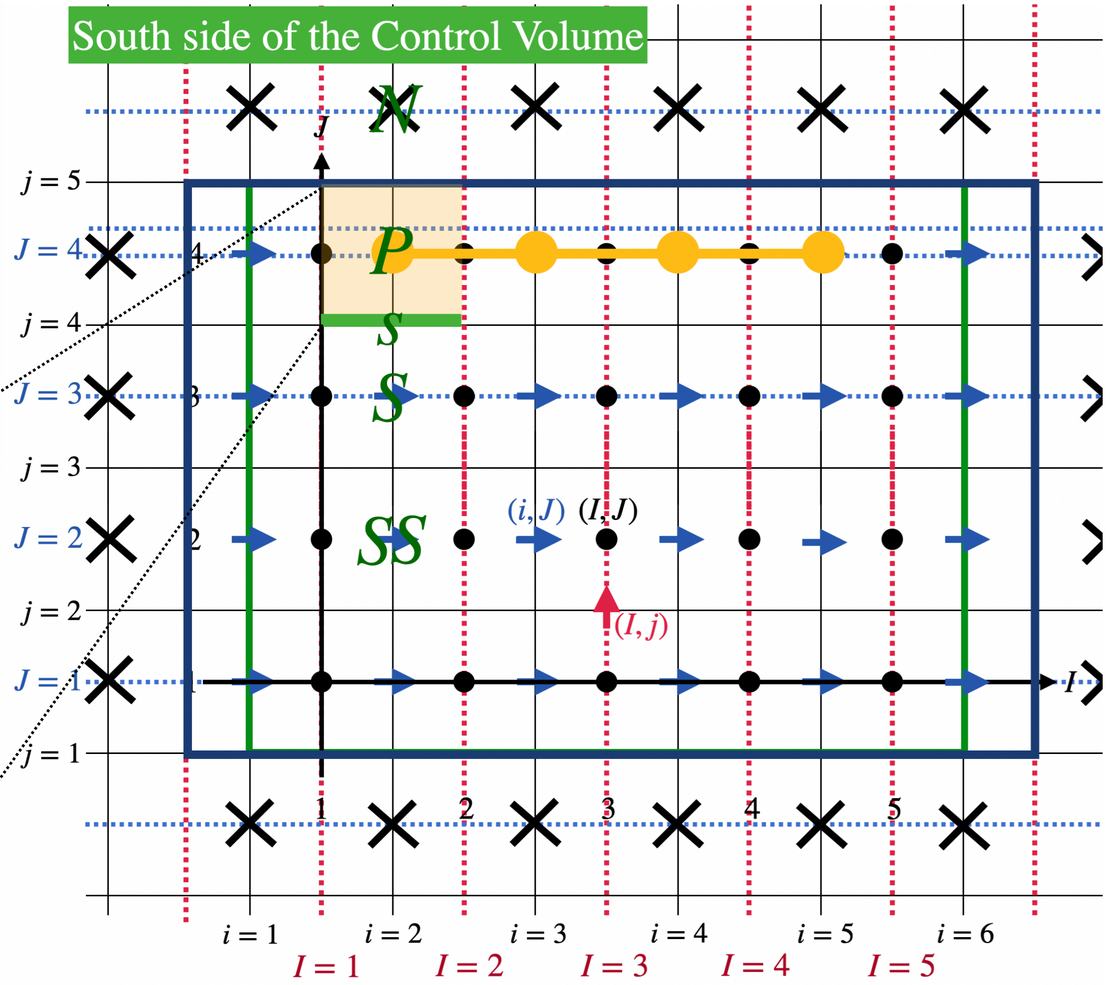
$$u_s = \left(\alpha_s UU(r_s)\right) u_{SS} + \left((1-\alpha_s)D(r_s) + \alpha_s U(r_s)\right) u_S$$

$$+\left((1-\alpha_s)U(r_s) + \alpha_s D(r_s)\right) u_P + \left((1-\alpha_s)UU(r_s)\right) u_N$$

Where, $u_N = 2u_{top} - u_P$



▲ Fig. 3.14 The control volume of the top boundary points



▲ Fig. 3.15 For south side treatment, j = Ny grids require special consideration for convection term !

3. Boundary Treatment : Summary

- Part III.Boundary treatment structure - Taking u-velocity as an example

Side	Convection term	Diffusion term	Treatment
West	$i = 2$		$Upwind \ : \quad u_w = (\alpha_w)u_{left} + (1-\alpha_w)u_P \quad *$
East	$i = Nx - 1$		$Upwind \ : \quad u_e = (\alpha_e)u_P + (1-\alpha_e)u_{right} \quad *$
South	$j = 1$	$j = 1$	$Expolation \ : \ u_S = 2u_{bottom} - u_P \qquad Given \ : \ u_s = u_{bottom}$
	$j = 2$		$Expolation \ : u_{SS} = 2u_{bottom} - u_S$
	$j = Ny$		$Expolation \ : \ u_N = 2u_{top} - u_P$
North	$j = 1$		$Expolation \ : \ u_S = 2u_{bottom} - u_P$
	$j = Ny - 1$		$Expolation \ : u_{NN} = 2u_{bottom} - u_N$
	$j = Ny$	$j = Ny$	$Expolation \ : \ u_N = 2u_{top} - u_P \qquad Given \ : \ u_n = u_{top}$
i = 1 & i = Nx			$u_P = B.C.$

* Where, if $F_w^u > 0$, $\alpha_w > 0$, else $\alpha_w < 0$. And if $F_e^u > 0$, $\alpha_e > 0$, else $\alpha_e < 0$

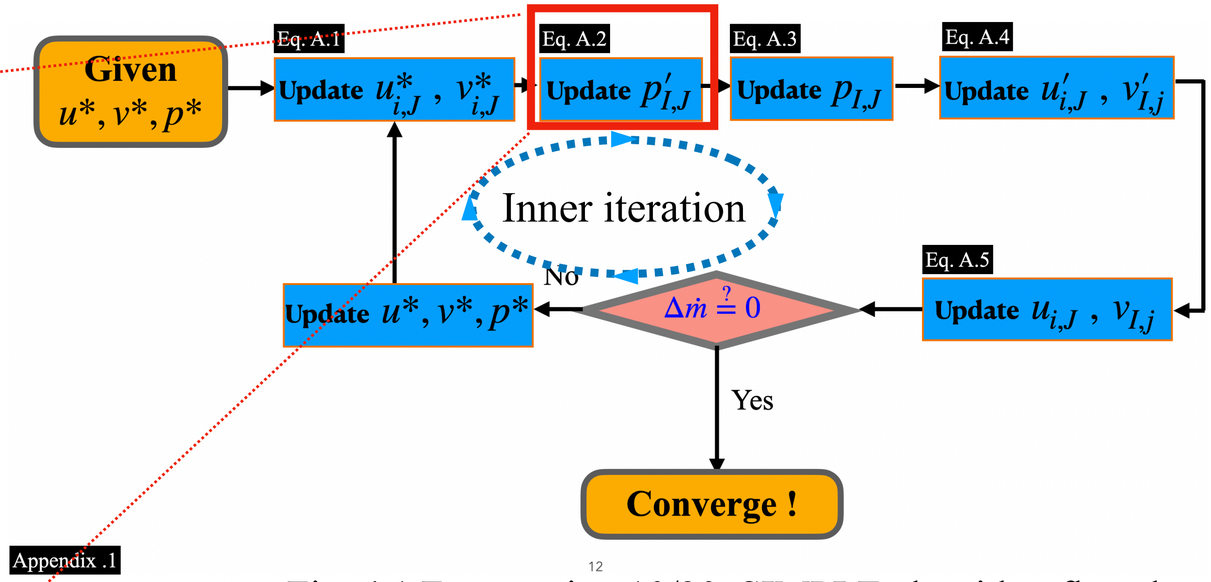
4. Convergence Criteria

1. Inner iteration :

(1) Equation : discrete pressure poisson equation

Poisson equation : $\nabla^2 p = f(\vec{r})$

Discrete poisson equation : Eq. (4.1)



▲ Fig. 4.1 For meeting 10/30, SIMPLE algorithm flowchart

$$\left(\frac{\delta y}{\rho(a_P)^{*\,N+\frac{1}{2}}_{i+1,J}}+\frac{\delta y}{\rho(a_P)^{*\,N+\frac{1}{2}}_{i,J}}+\frac{\delta x}{\rho(b_P)^{*\,N+\frac{1}{2}}_{I,j+1}}+\frac{\delta x}{\rho(b_P)^{*\,N+\frac{1}{2}}_{I,j}}\right)p'^{\,N+1}_{I,J}=\left(\frac{\delta x}{\rho(b_P)^{*\,N+\frac{1}{2}}_{I,j}}\right)p'^{\,N}_{I,J-1}+\left(\frac{\delta x}{\rho(b_P)^{*\,N+\frac{1}{2}}_{I,j+1}}\right)p'^{\,N}_{I,J+1}+\left(\frac{\delta y}{\rho(a_P)^{*\,N+\frac{1}{2}}_{i,J}}\right)p'^{\,N}_{I-1,J}+\left(\frac{\delta y}{\rho(a_P)^{*\,N+\frac{1}{2}}_{i+1,J}}\right)p'^{\,N}_{I+1,J}$$

$$+u^{*\,N+\frac{1}{2}}_{i,J}-u^{*\,N+\frac{1}{2}}_{i+1,J}+v^{*\,N+\frac{1}{2}}_{I,j}-v^{*\,N+\frac{1}{2}}_{I,j+1}=0 \qquad (4.1)$$

(2) Convergence Criteria : Gauss-Seidel convergence criteria and max iteration number = 1000.

$$\sum_i^N \frac{|f_i^{n+1}-f_i^n|}{\frac{|f_i^{n+1}|+|f_i^n|}{2}+q} \le 1e-5 \qquad (4.2)$$

4. Convergence Criteria

2. Outer iteration : SIMPLE algorithm

I have tried 2 types of criteria of otter iteration :

Define the variable $d_{I,J}$ R_{avg} :

$$d_{I,J} = A_e u_{i,J}^{*\,N+\frac{1}{2}} - A_w u_{i+1,J}^{*\,N+\frac{1}{2}} + A_n v_{I,j}^{*\,N+\frac{1}{2}} - A_s v_{I,j+1}^{*\,N+\frac{1}{2}}$$

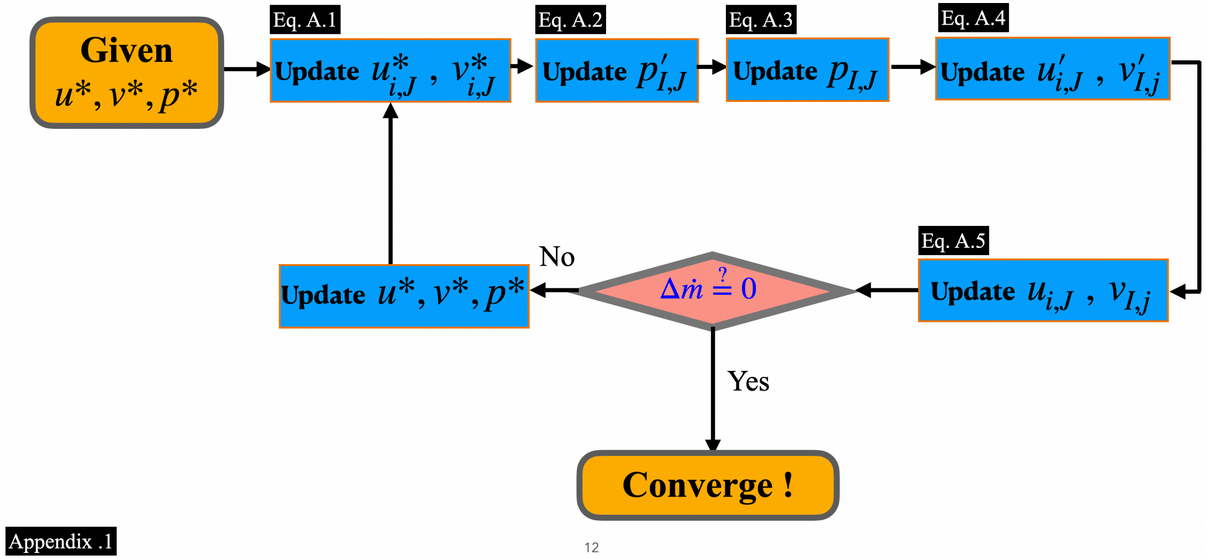
$$R_{avg} = \sum_{i}^{N} \frac{|d_{I,J}|}{N}$$,and I called “average flow rate”

(1)Relative error of the average flow rate:

$$\frac{R_{avg}^{n+1} - R_{avg}^{n}}{R_{avg}^{n+1}} \leq 1e-5 \qquad (4.3)$$

(2) Average flow rate :

$$R_{avg} \leq 1e-10 \qquad (4.4)$$

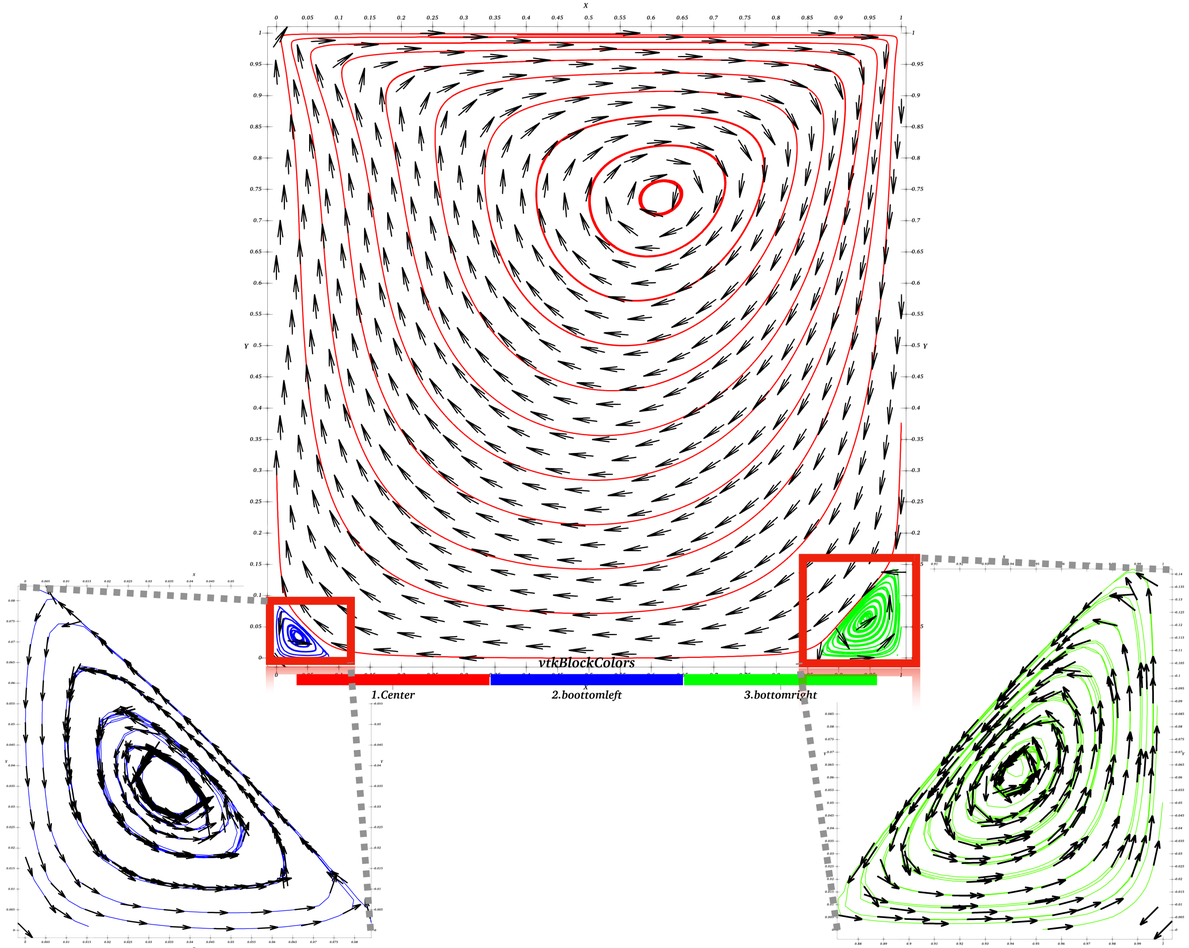


▲ Fig. 4.1 For meeting 10/30, SIMPLE algorithm flowchart

5. Result and Discussion

(1) Visualize Velocity Vector Field and Streamlines : Comparison at Re = 100 and Re = 1000 for QUICK

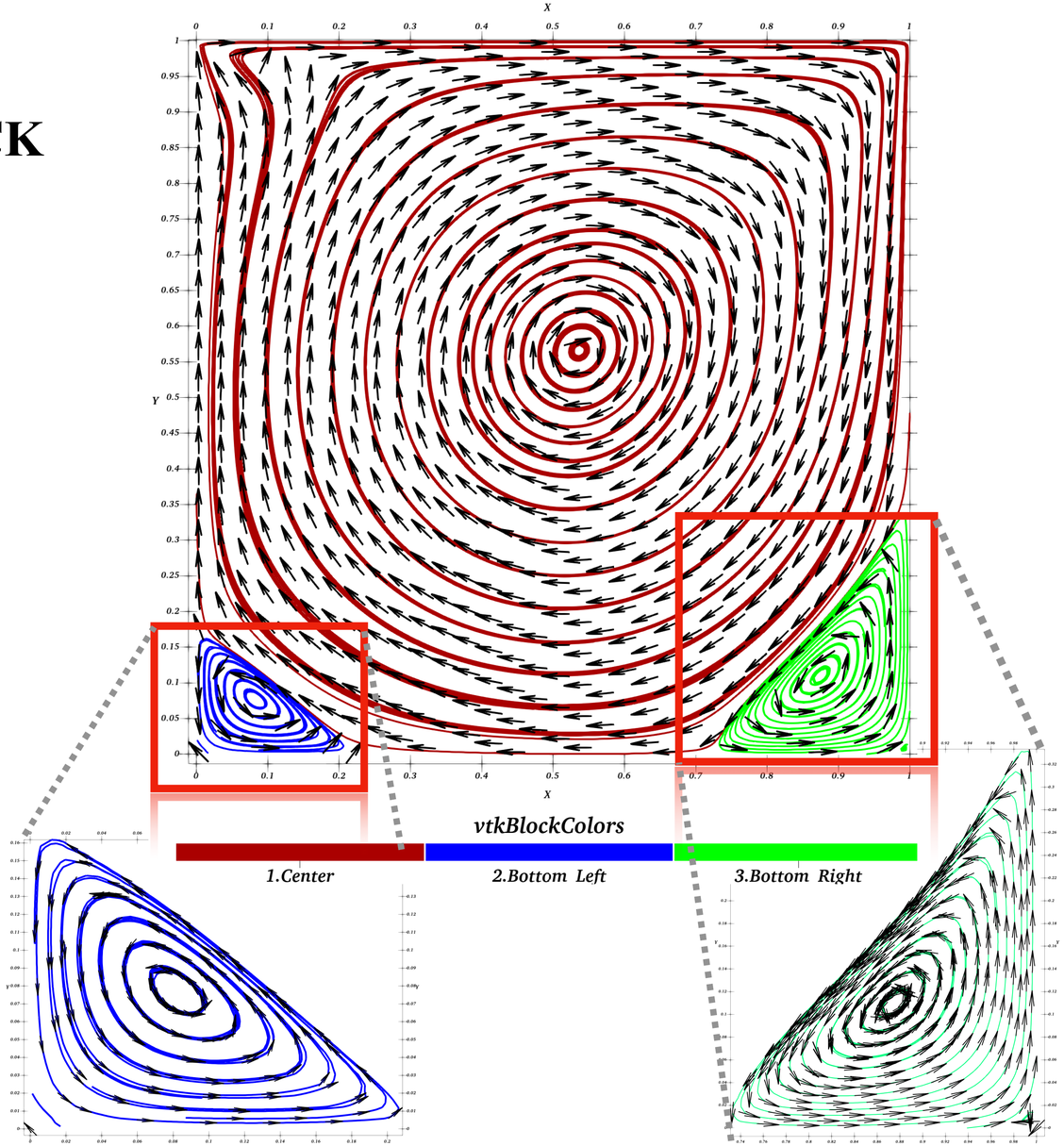
$Re = 100$ $Grid$ 161×161



▲ Fig. 5.1 bottom-left vortex at Re = 100

▲ Fig. 5.2 bottom-right vortex at Re = 100

$Re = 1000$ $Grid$ 161×161



▲ Fig. 5.3 bottom-left vortex at Re = 1000

▲ Fig. 5.4 bottom-right vortex at Re = 1000

5. Result and Discussion

(1) Visualize Velocity Vector Field and Streamlines : Re = 5000 for QUICK scheme

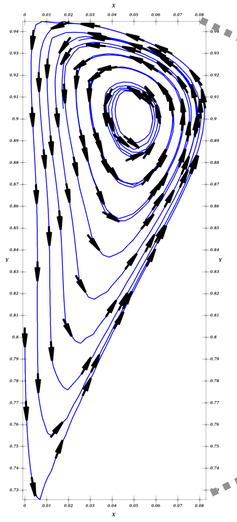


Fig. 5.4 top-left vortex at Re = 5000

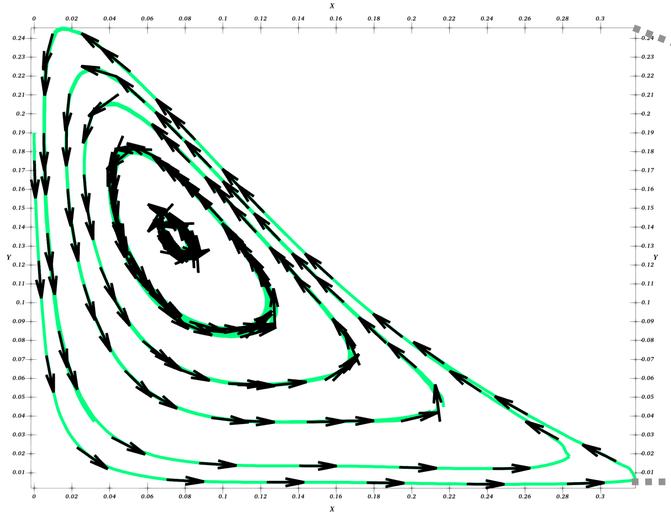
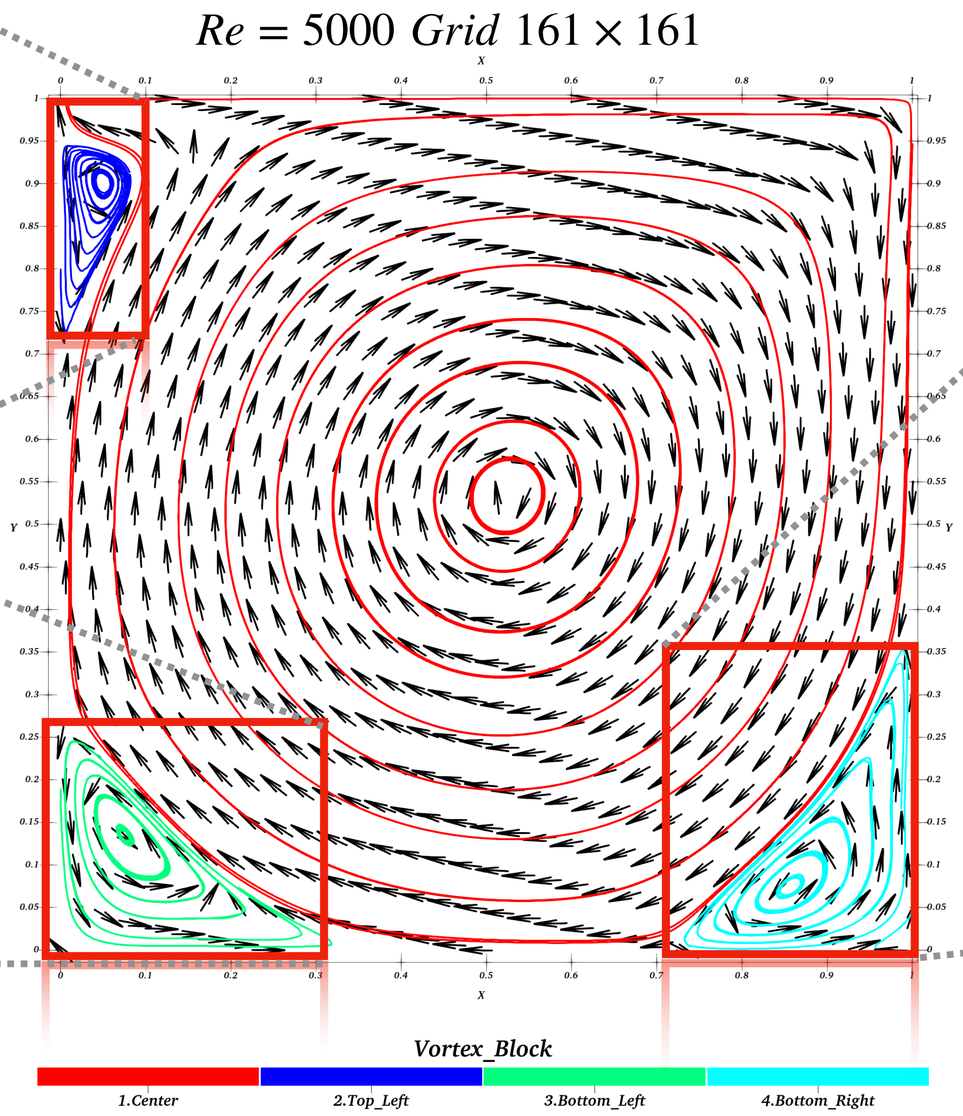


Fig. 5.5 bottom-left vortex at Re = 5000



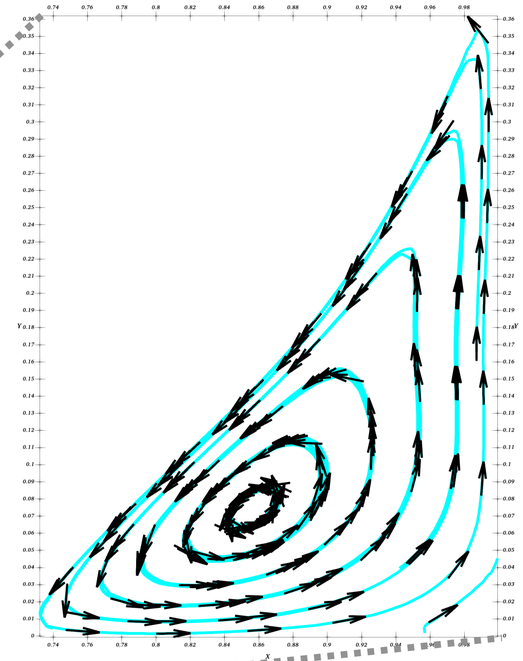
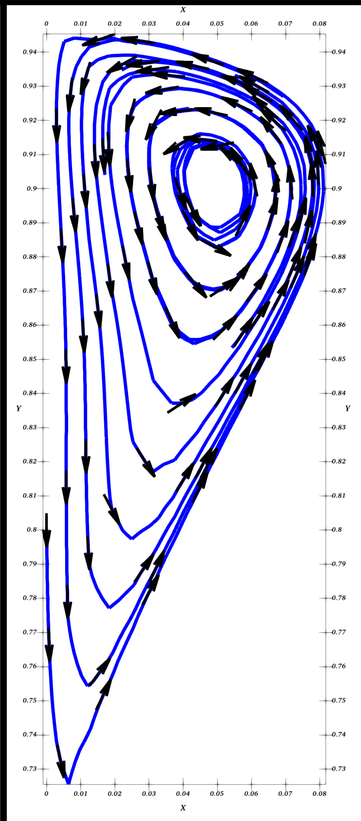
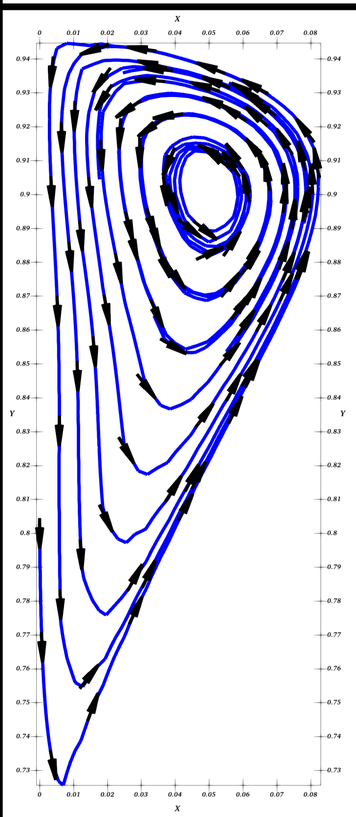
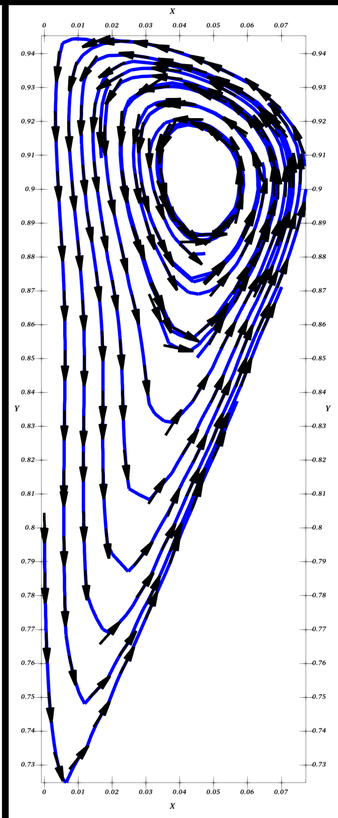


Fig. 5.6 bottom-right vortex at Re = 5000

5. Result and Discussion

(1) Visualize Velocity Vector Field and Streamlines :

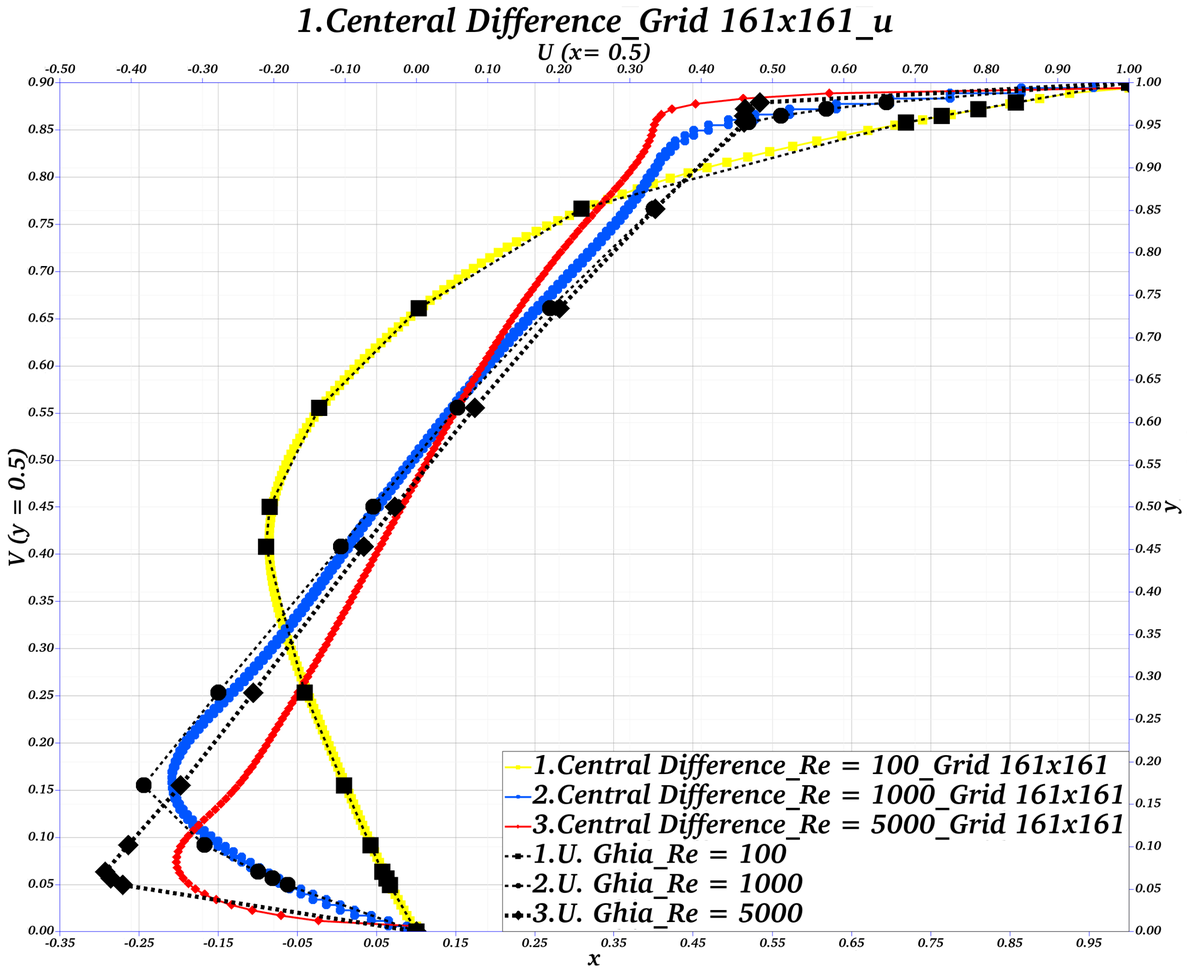
- Top-Left Secondary Vortex Size: Comparison of CD, QUICK, and TVD-MUSCL Schemes

Scheme	Central Difference	QUICK	TVD-MUSCL
Top-Left Vortex	 Vortex Range in x-direction : 0 ~ 0.08	 Vortex Range in x-direction : 0 ~ 0.08	 Vortex Range in x-direction : 0 ~ 0.07

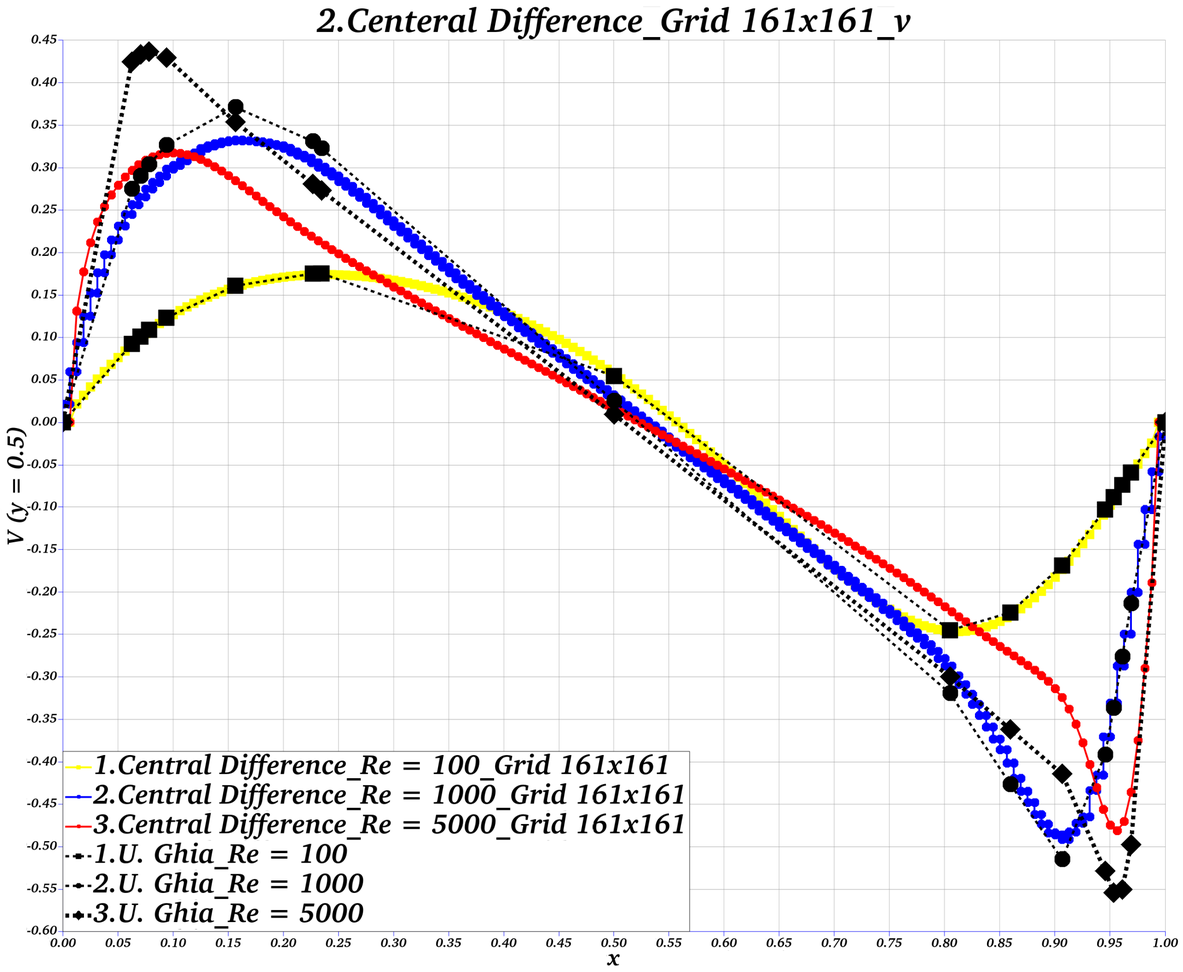
▲ Fig. 5.7 CDs_Re = 5000_Grid 161x161 ▲ Fig. 5.8 QUICK_Re = 5000_Grid 161x161 ▲ Fig. 5.9 TVD-MUSCL_Re = 5000_Grid 161x161

5. Result and Discussion

(2) Compare centerline velocity profiles with different Re numbers : CDs



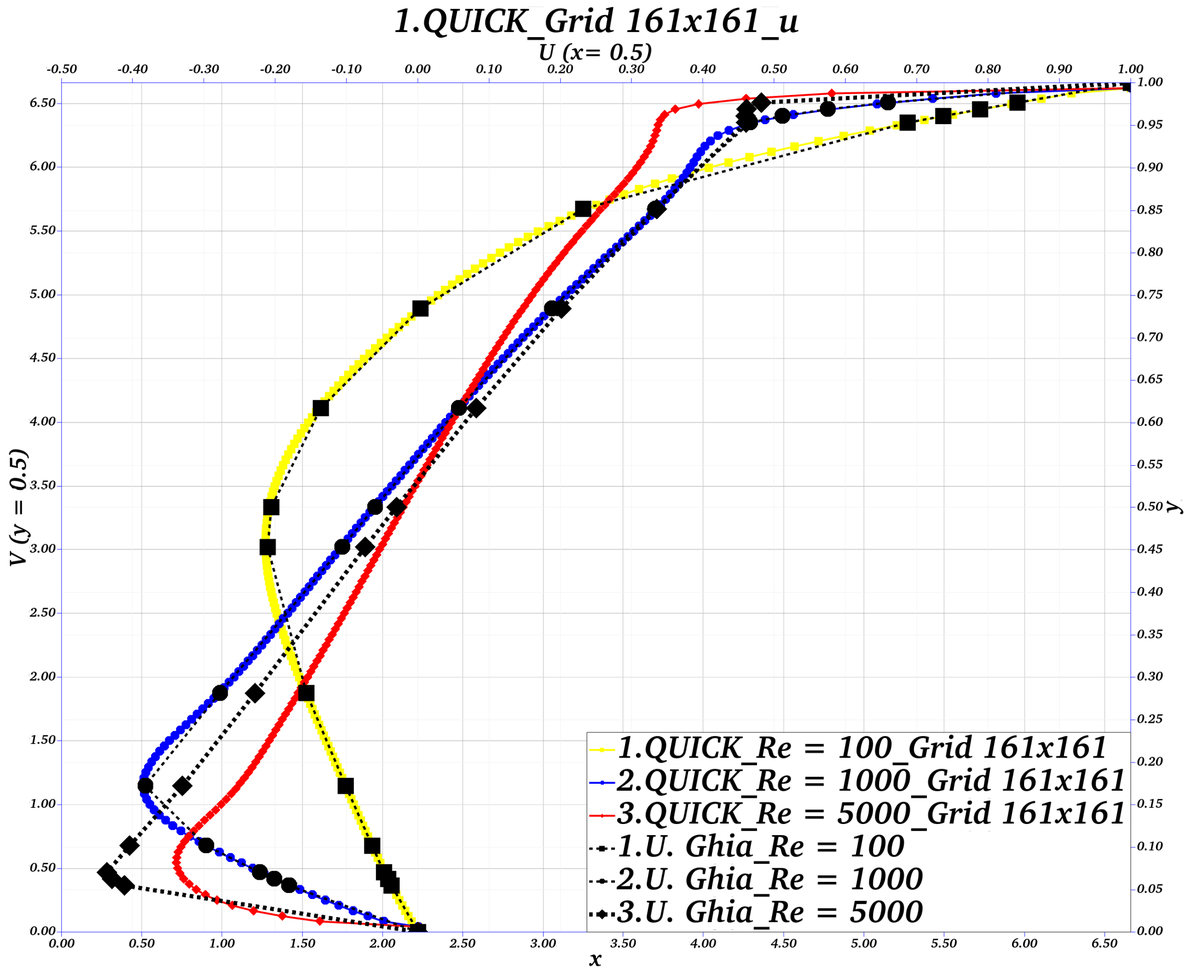
▲ Fig. 5.10 CDs_Re = 5000_Grid 161x161_u-component



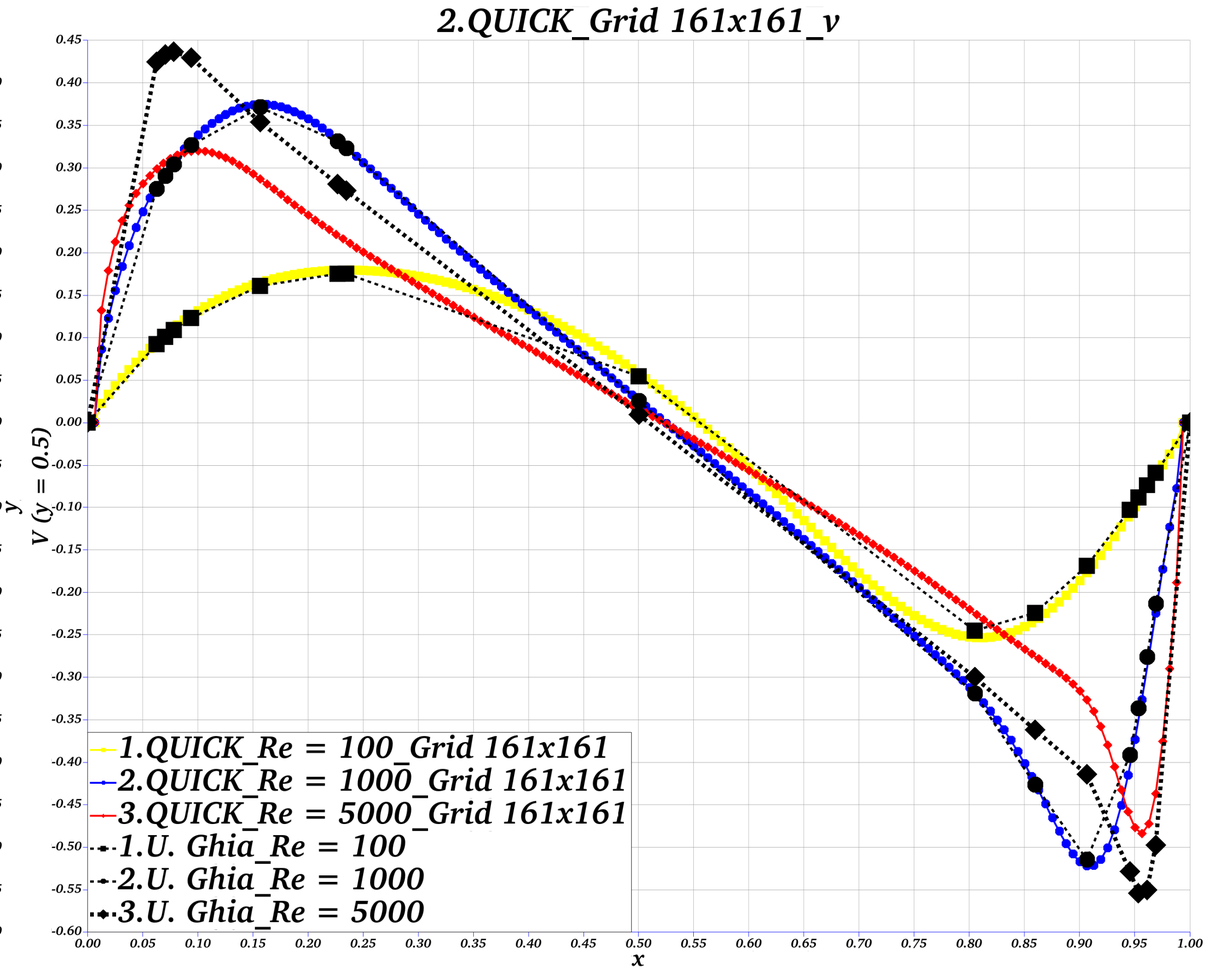
▲ Fig. 5.11 CDs_Re = 5000_Grid 161x161_v-component

5. Result and Discussion

(2) Compare centerline velocity profiles with different Re numbers : QUICK



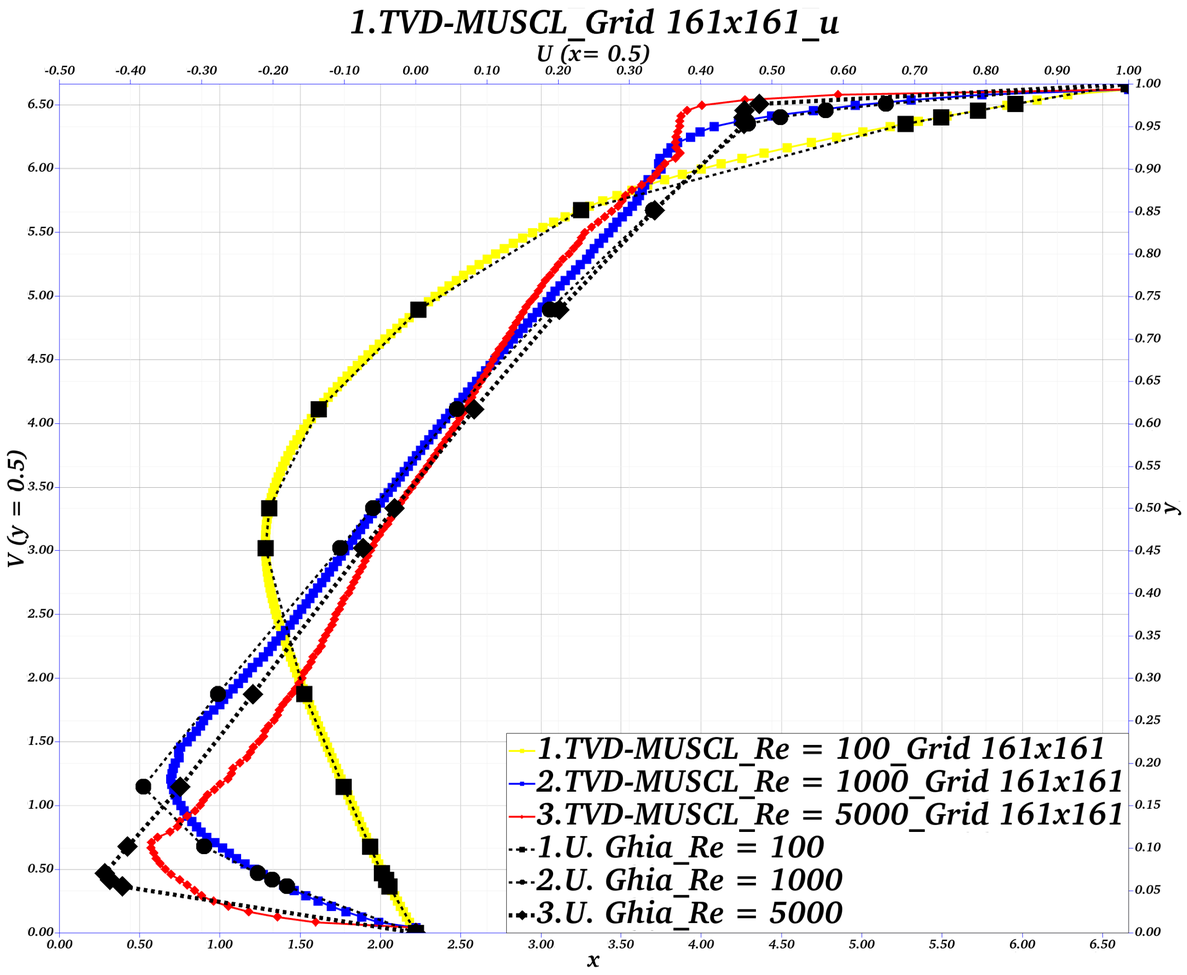
▲ Fig. 5.12 QUICK_Re = 5000_Grid 161x161_u-component



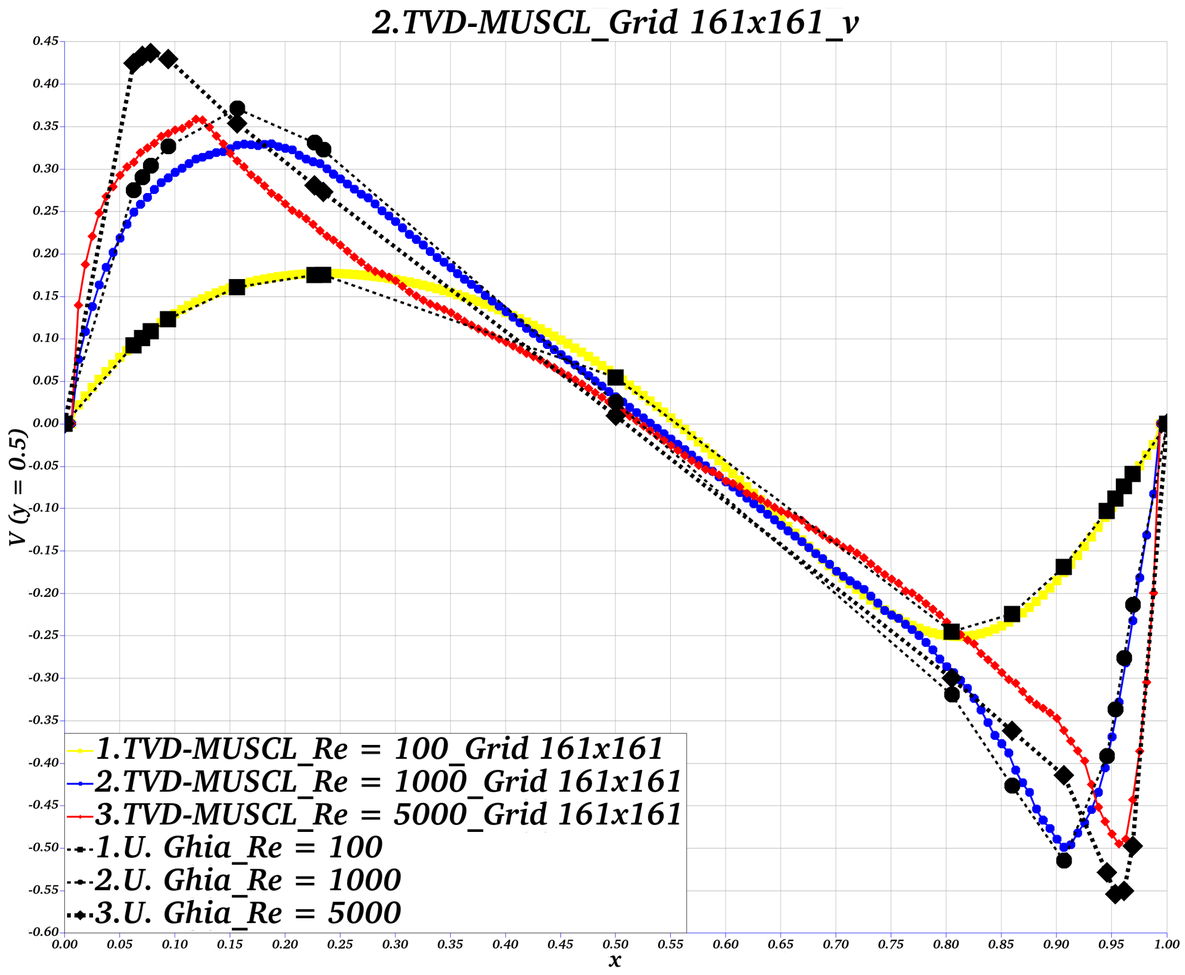
▲ Fig. 5.13 QUICK_Re = 5000_Grid 161x161_v-component

5. Result and Discussion

(2) Compare centerline velocity profiles with different Re numbers : TVD-MUSCL



▲ Fig. 5.14 TVD_MUSCL_Re = 5000_Grid 161x161_u-component

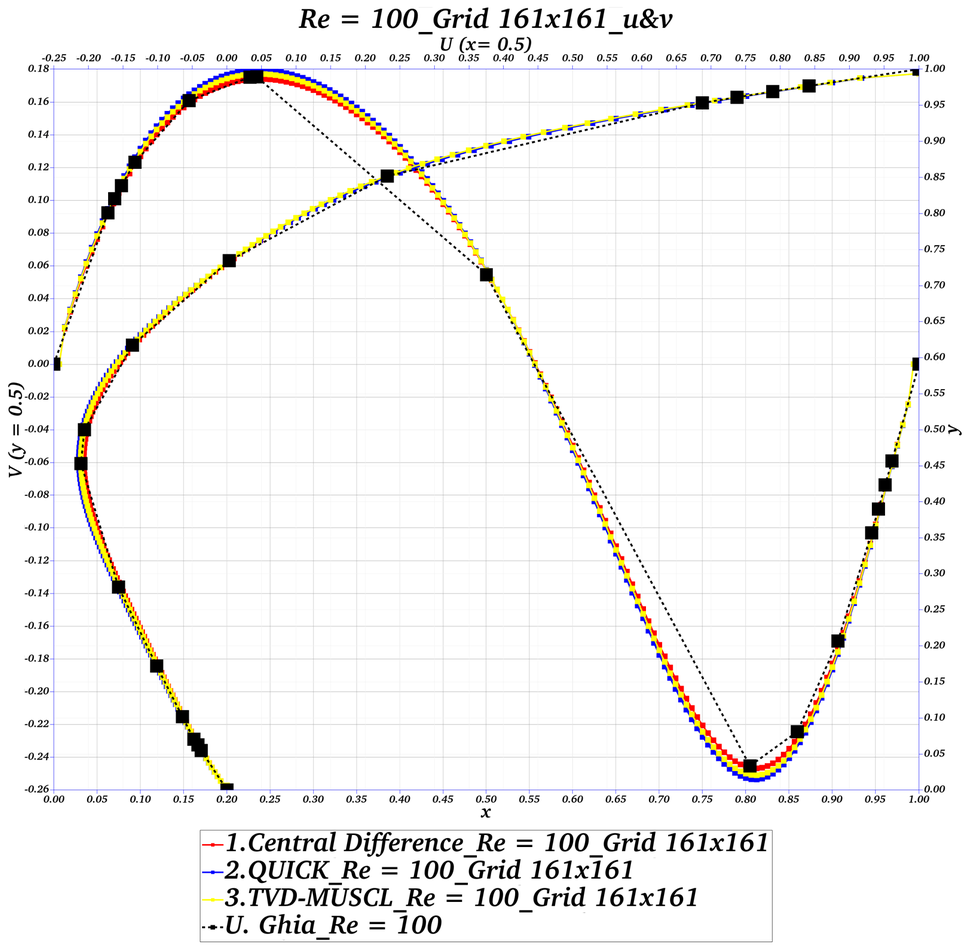


▲ Fig. 5.15 TVD_MUSCL_Re = 5000_Grid 161x161_v-component

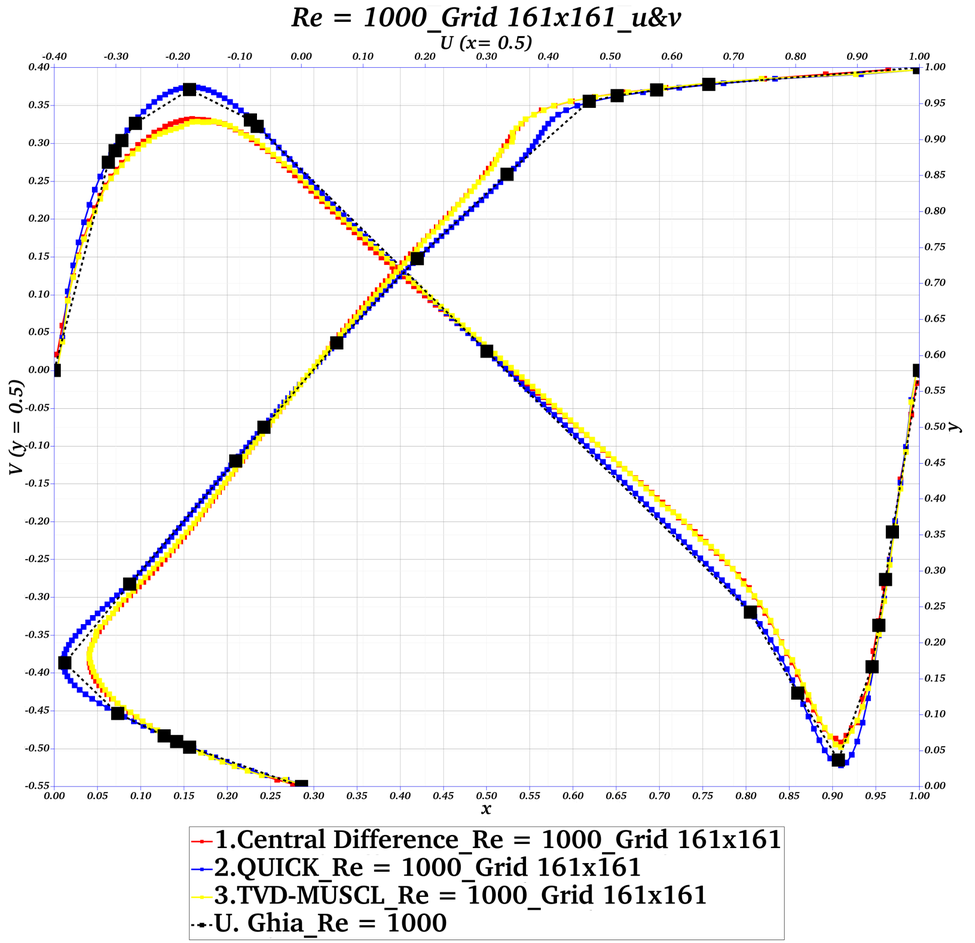
5. Result and Discussion

(3) Different compare centerline velocity profiles with different numerical schemes

- The centerline velocity profiles at Re = 100 and 1000 closely match the benchmark data.



▲ Fig. 5.16 Centerline velocity profiles (Re = 100 , Grid 161x161)

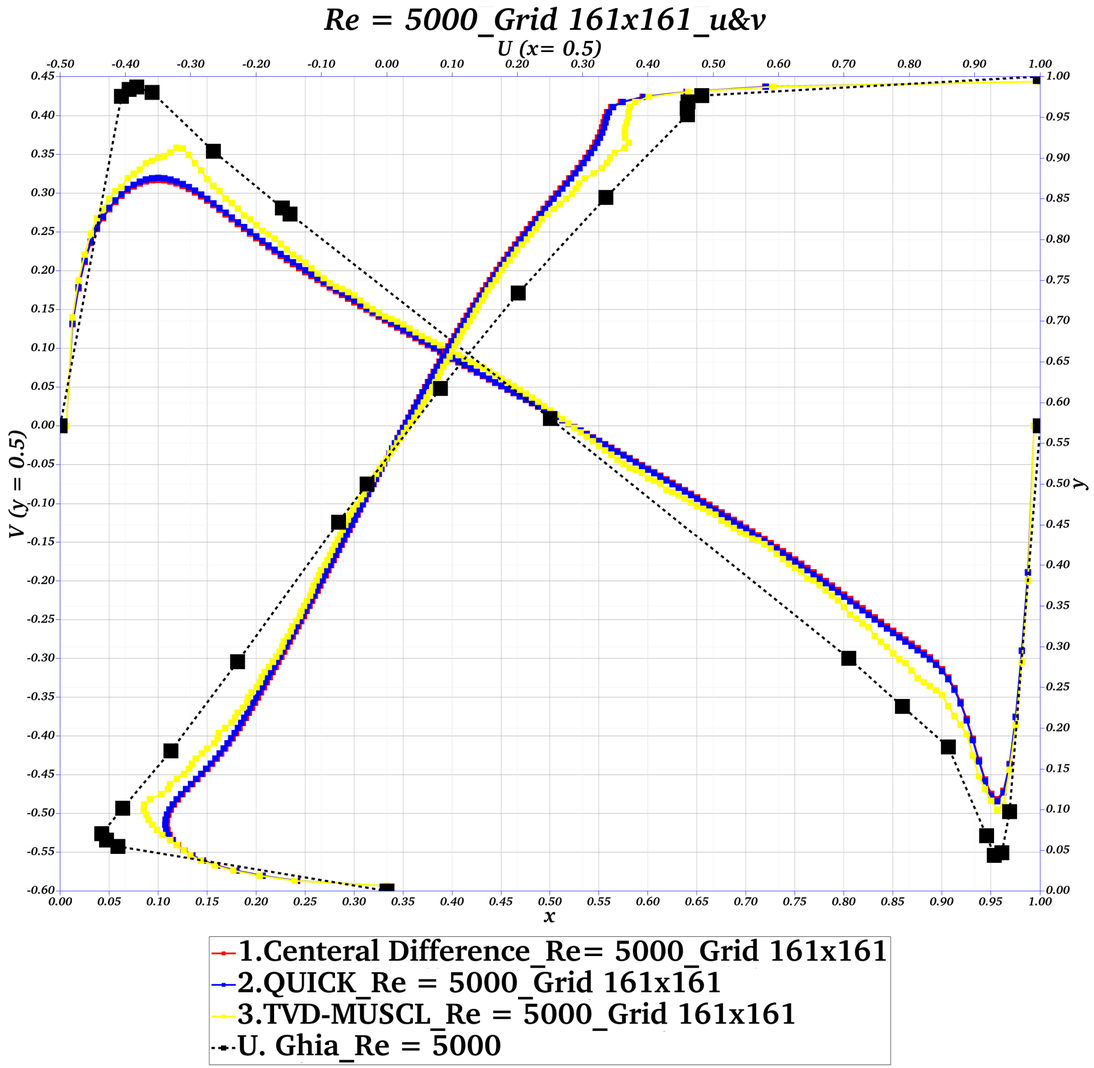


▲ Fig. 5.17 Centerline velocity profiles (Re = 1000 , Grid 161x161)

5. Result and Discussion

(3) Different compare centerline velocity profiles with different numerical schemes

- The centerline velocity profiles at Re = 5000

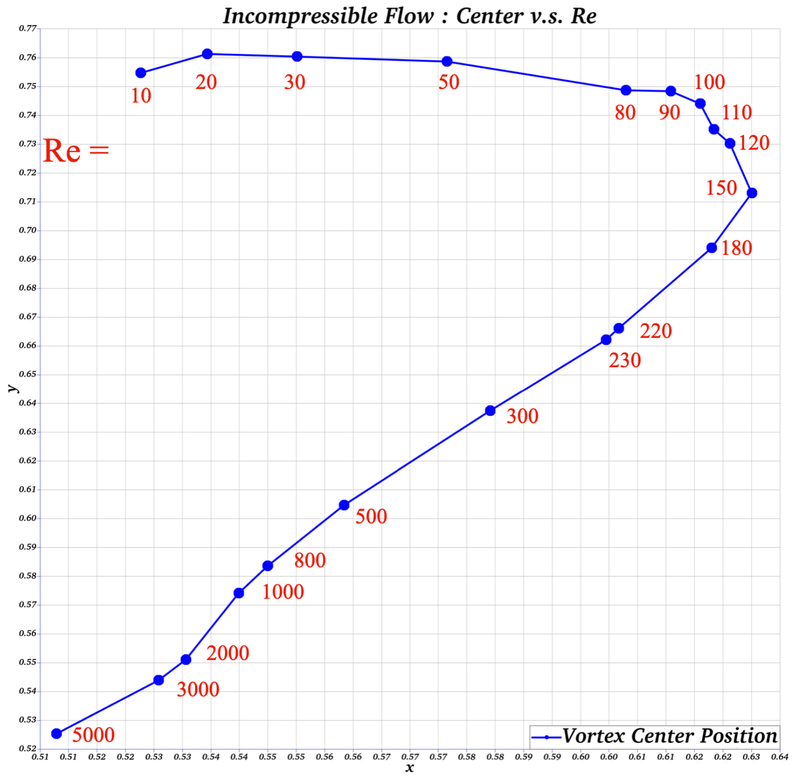


6. Conclusion

1.My challenge: From Re = 1000 to 5000, I tried progressively smaller relaxation factors, but most choices failed to converge. After correcting my code and implementing the deferred correction method, the convergence improved significantly. However, I am concerned about whether this method affects the spatial accuracy of the QUICK and TVD-MUSCL schemes.

2.Using Lattice Boltzmann Method :

Find the variation in vortex center position for Reynolds numbers ranging from Re = 10 to 5000



▲ Fig. 6.1 vortex center position v.s. Re